12 - Experimental Results and Conclusion

Only numbers produced by the pipeline

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

models/model_metadata/last_run_summary.json; models/evaluation/*; report/tables/*; report/figures/*

Setup

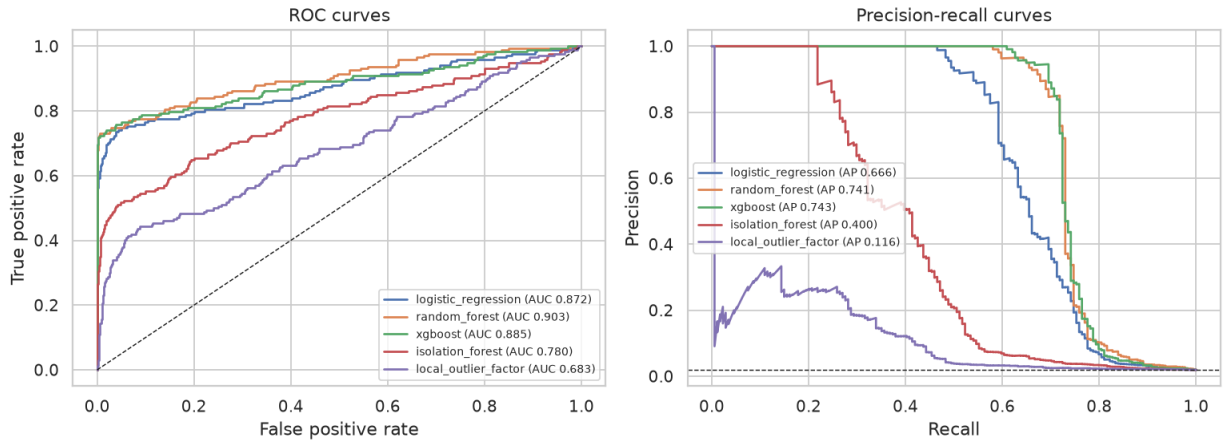
Item	Value
Run id	run_20261109_061947_57b9
Seed	42
Rows after cleaning	59682
Fraud rate	2.0207%
Tuning iterations per model	8
Library versions	{'python': '3.11.18', 'sklearn': '1.9.1', 'xgboost': '3.2.0', 'pandas': '3.1.1', 'numpy': '2.4.6'}
Total pipeline seconds	245.3

Model comparison

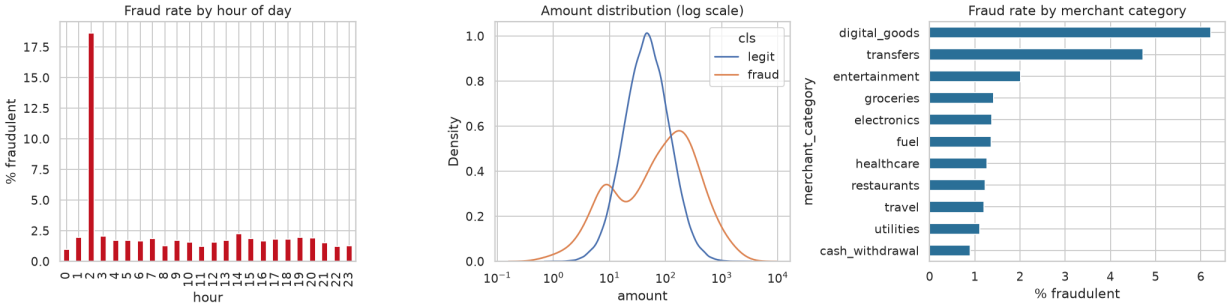
Model	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR
logistic_regression	0.883	0.563	0.688	0.872	0.666	0.0015
random_forest	0.928	0.667	0.776	0.903	0.741	0.0010
xgboost	0.982	0.626	0.765	0.885	0.743	0.0002
isolation_forest	0.365	0.437	0.398	0.780	0.400	0.0150
local_outlier_factor	0.187	0.287	0.226	0.683	0.116	0.0248

Champion (highest PR-AUC among supervised models): xgboost. Fraud cases in the test set: 174; with so few positives the metrics have wide uncertainty - see the bootstrap interval [0.8471476461231401, 0.9181854459621215].

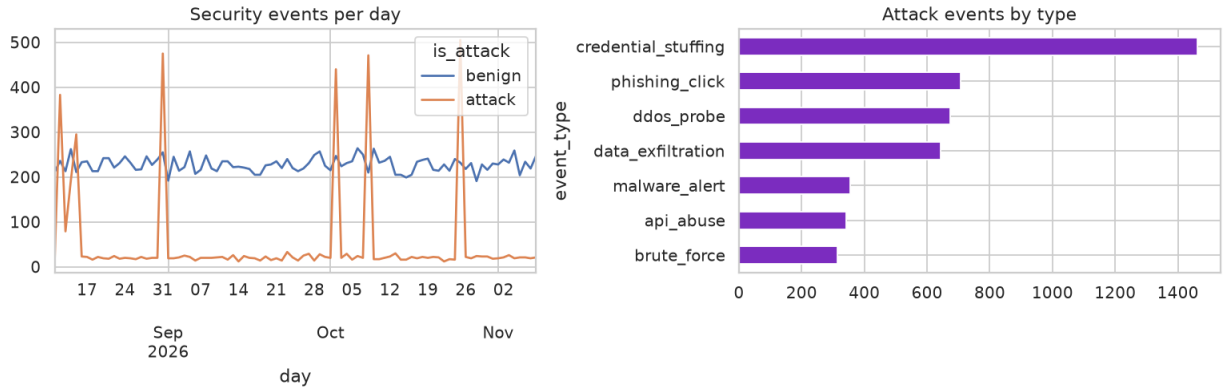
Image not available: model_comparison.png. ()



ROC and precision-recall curves.



Transaction overview: fraud by hour, amount distribution, fraud by category.



Security events overview.

Interpretation guardrails

- Data are synthetic; fraud patterns were designed by the authors, so high scores show the method works on that design, not that real fraud is this easy.
- Results come from a single seed and a small tuning budget; they are not averaged over repeats.
- The survey is synthetic; trust numbers are demonstration values.
- No claim is made about real-world deployment performance.

Conclusions the project can support

- A reproducible, containerised pipeline links fraud and anomaly detection with a consumer trust measure and monitors the experiment.
- Models are compared with imbalance-aware metrics and false-positive cost rather than accuracy alone.
- Unsupervised detectors are evaluated separately from supervised models.

- Dashboards, logs and database views are driven by computed data.

Limitations (from the project documents)

- Synthetic data encode assumptions; no concept drift.
- Synthetic survey; no ethics approval recorded.
- Trust Index is a proposal without psychometric validation.
- Replay stream is not production traffic.
- Alertmanager not deployed.

Future work (planned, not implemented)

- Run on the public ULB dataset and repeat across seeds.
- Real ethics-approved survey; factor analysis.
- Calibration, cost-sensitive thresholds, SHAP explanations.
- Drift monitoring, scheduled retraining, Alertmanager.
- Use unsupervised scores as features for supervised models.